To follow along in your own workspace, please open the following notebooks:

- `CH7-01-GetData`
- `CH7-02-Evaluating ChatBot`

Loading the ground truth labels

We start in the first notebook, `CH7-01-GetData`. To evaluate our model, we must have ground truth labels, a.k.a. the correct answers. This generally involves human effort to write out some typical questions you expect users to ask your chatbot and the answers you expect your chatbot to respond with. To simplify this process, we created a file with thirty-five sample questions and the corresponding human-generated answers for this project, saved to `Questions_Evaluation.csv`. Let's load this file and examine the question and answer pairs.

```
import pandas as pd

df_evaluation = pd.read_csv("/Volumes/ml_in_action/rag_chatbot/files/Questions_Evaluation.csv").dropna().
drop(columns=["id_question"])
df_eval = spark.createDataFrame(df_evaluation)
display(df_eval)
```

Figure 7.42 – Reading our pre-created evaluation set

Take a look at some of the records to get a sense of the questions a user might ask and the expected answers. You can also add your own examples to augment the evaluation data further.

question_asked	answer_given
How does the choice between rules and standards impact content moderation on social media platforms?	Rules can provide clear guidelines for content moderators to follow, reducing the need for human involvement an potentially lowering error costs. However, rules may not be effective in addressing complex or dynamic issues, su hate speech, and may require frequent updates to remain effective. Standards, on the other hand, can provide fle for content moderators to adjust their policies to specific contexts, but may require more human involvement and potentially increase error costs d...
What are large language models (LLMs) and how do they differ from other generative AI models?	Large language models (LLMs) are artificial intelligence models that can process and produce various forms of se data, including but not limited to natural language. They differ from other generative AI models as they can be tra using a range of architectures and are not limited to transformer-based models. LLMs can generate a wide range sequences, such as assembly language, protein sequences, and chess games.

Figure 7.43 – Question-answer pair examples

Let's save these examples to a Delta table named `evaluation_table`. That way, if a new `Question_Evaluation.csv` file with different examples is uploaded, you can append the new examples to the existing table without risking losing the original data.